

Renewable Energy Integration Project

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1 Introduction

This document is used to show the methodologies adopted during the development of the renewable energy integration project.

2 Project Goal

The project aims to develop and analyse a simplified AC power flow using IEEE bus electrical network samples. The system is being designed as a structured analysis/ engineering web application focused on:

- Load flow analysis,
- Integration of renewable energy sources,
- Optimisation of load flow.

The project is intended both as:

- a technical learning platform.
- professional development.

3 IEEE 5 Bus AC Load Flow analysis